

CBSE Practice Worksheet

Class 8 | Science | Difficulty: Hard | Set C

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. What is the oxidation state of Mn in KMnO_4 ?
(A) +2
(B) +4
(C) +6
(D) +7
2. What is the wavelength of de Broglie wave for an electron?
(A) $\lambda = h/mv$
(B) $\lambda = mv/h$
(C) $\lambda = hv/m$
(D) $\lambda = m/hv$
3. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
(A) 25%
(B) 33%
(C) 50%
(D) 75%
4. Which principle explains the working of a hydraulic lift?
(A) Archimedes
(B) Pascal
(C) Bernoulli
(D) Newton
5. Which type of mutation involves change in single nucleotide?
(A) Deletion
(B) Insertion
(C) Point mutation
(D) Frameshift
6. What is the escape velocity from Earth's surface?
(A) 8 km/s
(B) 11.2 km/s
(C) 15 km/s
(D) 25 km/s
7. Which quantum number determines the shape of an orbital?
(A) Principal
(B) Azimuthal
(C) Magnetic
(D) Spin
8. Which reaction is endothermic?
(A) Combustion
(B) Neutralization
(C) Photosynthesis

(D) Respiration

9. What is the magnetic quantum number for 3d orbital?

(A) -2 to +2

(B) -3 to +3

(C) -1 to +1

(D) 0 only

10. Which organelle is involved in programmed cell death?

(A) Lysosome

(B) Mitochondria

(C) Peroxisome

(D) Ribosome

Section B: Fill in the Blanks (1 mark each)

11. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.

12. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.

13. The Krebs cycle occurs in the ____ of mitochondria.

14. In $\text{S}_\text{N}2$ reactions, the rate depends on both ____ and nucleophile.

15. The uncertainty principle was given by ____.

16. For a reversible process, entropy change of the universe is ____.

17. The process of splitting water during photosynthesis is called ____.

18. The enzyme that converts prothrombin to thrombin is ____.

Section C: Short Answer Questions (2 marks each)

19. Derive the lens formula and explain its applications.

20. Explain the structure and working of a nuclear reactor.

21. Explain Le Chatelier's principle with examples.

22. Derive the equation for kinetic energy of gases from kinetic theory.

23. Describe the light-dependent and light-independent reactions of photosynthesis.

24. Explain the mechanism of DNA replication.

25. Explain hybridization with examples of sp , sp^2 , and sp^3 .

--- End of Worksheet ---

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